

Communication (packed bytes)

| Object            | Notation       | Bytes | In the basic signature?     |
|-------------------|----------------|-------|-----------------------------|
| public key        | $pk = t$       | 4416  | yes (shared)                |
| secret key        | $sk = r$       | 704   | yes (shared)                |
| challenge         | $c$            | 64    | yes                         |
| response          | $z / \hat{z}$  | 6688  | yes                         |
| signature         | $(c, z)$       | 6752  | yes (basic signature)       |
| statement         | $Y = t'$       | 4416  | LAS only (public)           |
| witness           | $r'$           | 704   | LAS only (private)          |
| pre-signature     | $(c, \hat{z})$ | 6752  | LAS only                    |
| adapted signature | $(c, z)$       | 6752  | LAS (verifies as basic sig) |

Computation (µs/op)

| Operation          | basic (µs)    | LAS adaptor (µs)     | overhead   |
|--------------------|---------------|----------------------|------------|
| KeyGen             | $114 \pm 5$   | $114 \pm 5$ (shared) | —          |
| Sign / PreSign     | $1270 \pm 28$ | $1355 \pm 62$        | +6.7%      |
| Verify / PreVerify | $280 \pm 7$   | $288 \pm 2$          | +3.1%      |
| Verify / Adapt     | $280 \pm 7$   | $302 \pm 5$          | +8.1%      |
| Extract            | —             | $101 \pm 3$          | (LAS only) |